

DBMS LAB

ER FORENSICS

Assignment - 1

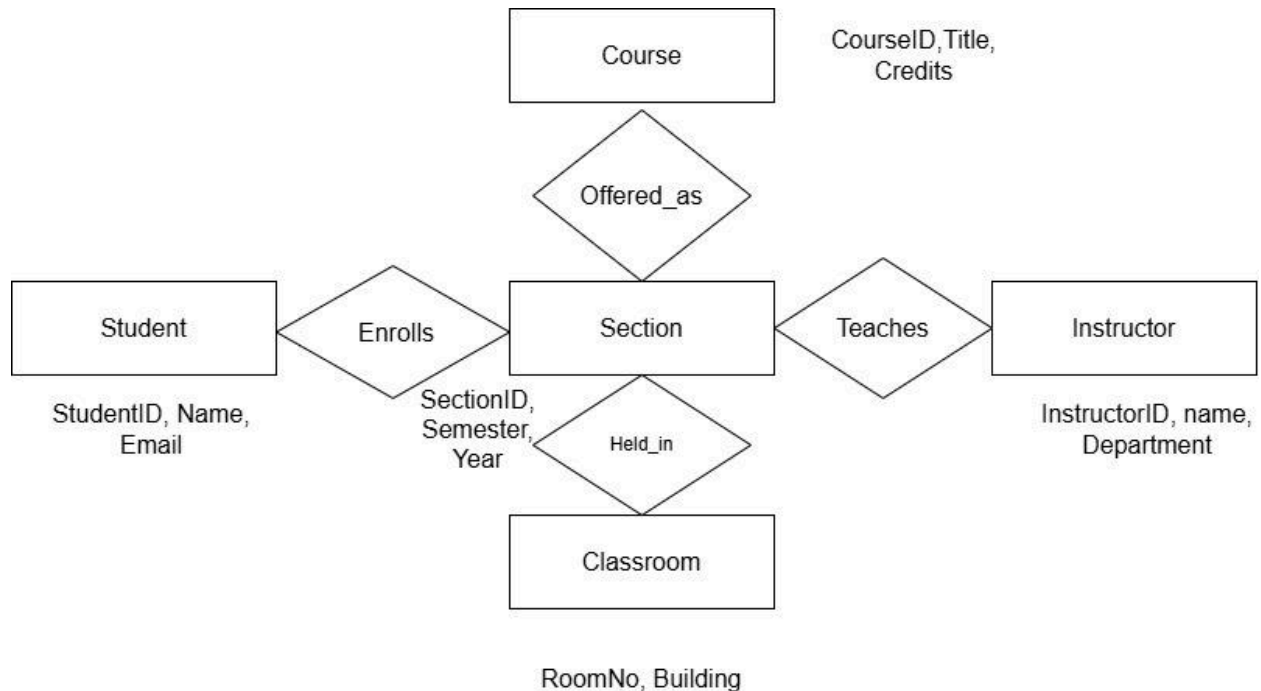
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Course: DBMS Lab

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1. University Course Registration

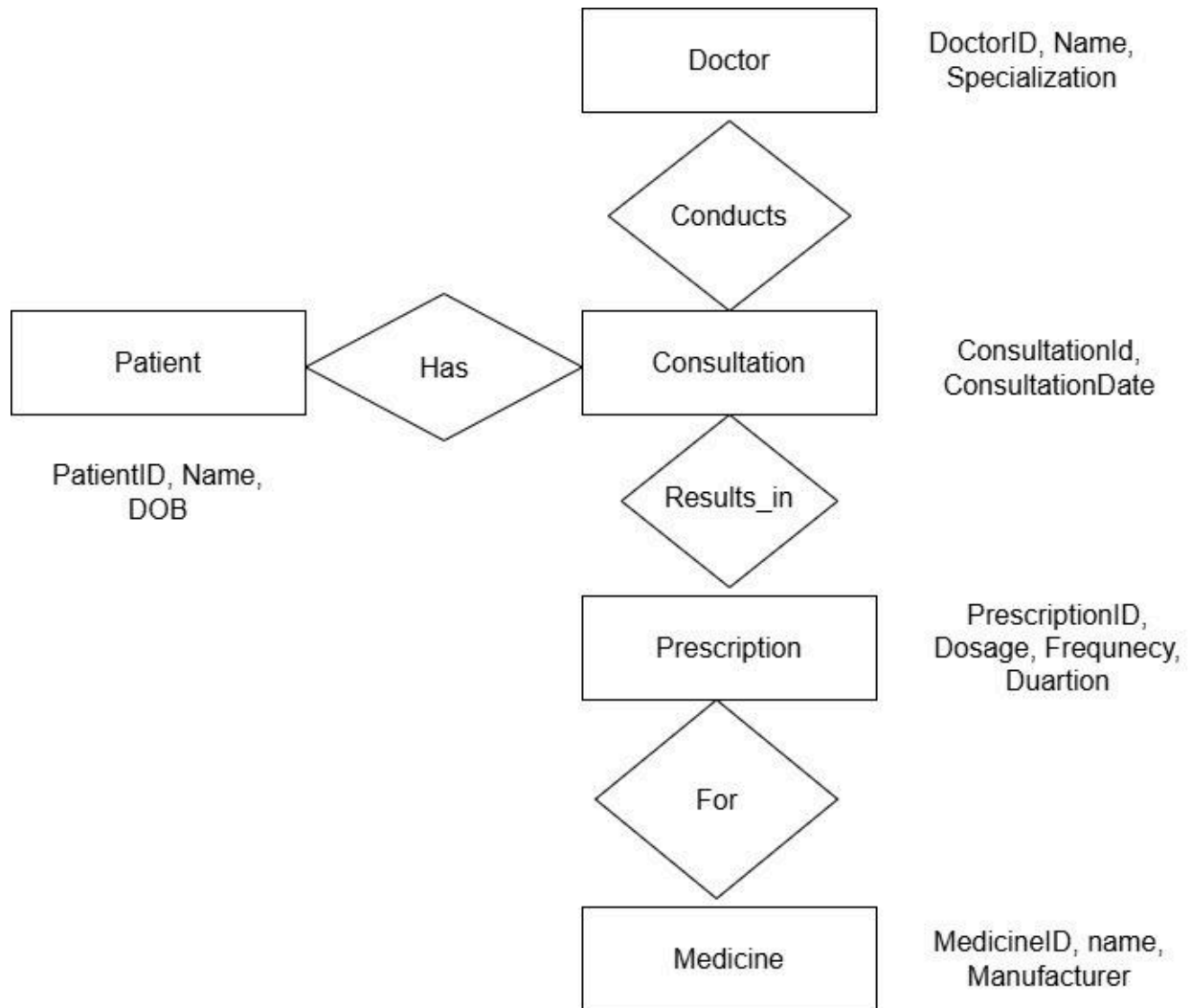
Corrected ER Diagram



The original ER model directly connected Student with Course, which could not represent that the same course may have multiple sections in the same semester and that a student registers for a particular section. To correct this, a **Section** entity was introduced between Course and Student, with Semester and Year information. The ENROLLS relationship was changed to connect Student with Section so that the exact section registered by a student is recorded. Similarly, TEACHES was changed to connect Instructor with Section because different sections of the same course may have different instructors. A HELD_IN relationship was also connected to Section so that different sections can be assigned different classrooms. These changes preserve the section-wise, instructor-wise, and classroom-wise information required by the business rules.

2. Hospital Prescription System

Corrected ER Diagram

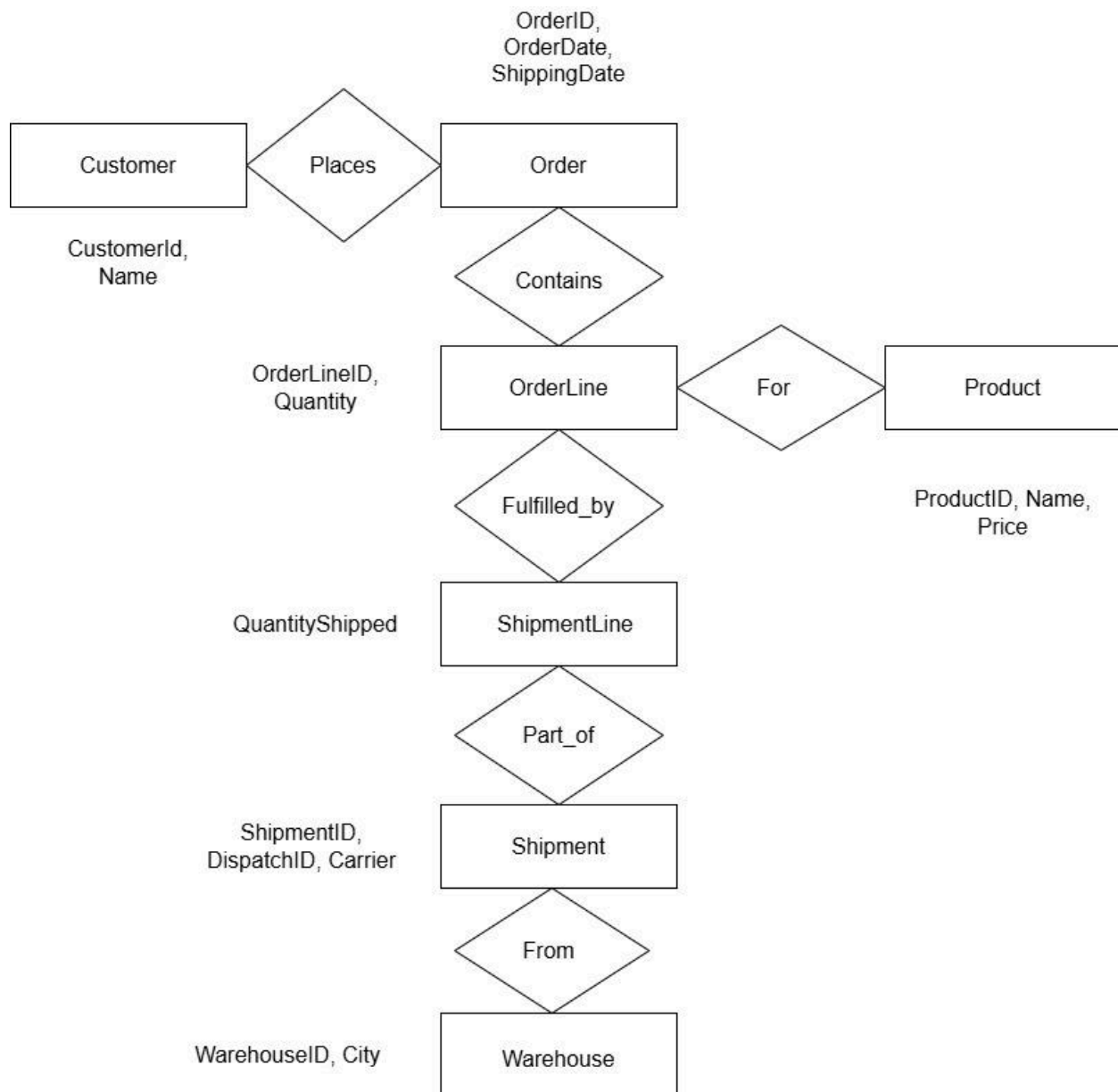


The original ER model could not distinguish between repeated consultations of the same patient with the same doctor or associate a prescription with a particular visit. To correct this, a **Consultation** entity was introduced to represent each individual visit, connecting Patient and Doctor through separate relationships. A **Prescription** entity was also introduced and associated with the Consultation so that each prescription can be linked to a specific visit. Dosage, frequency, and duration were placed with the Prescription rather than Medicine because the same

medicine can have different instructions for different patients and visits. This also allows the same medicine to be prescribed multiple times while preserving the complete prescription history and identifying who prescribed what, to whom, and when.

3. E-Commerce Order Fulfilment

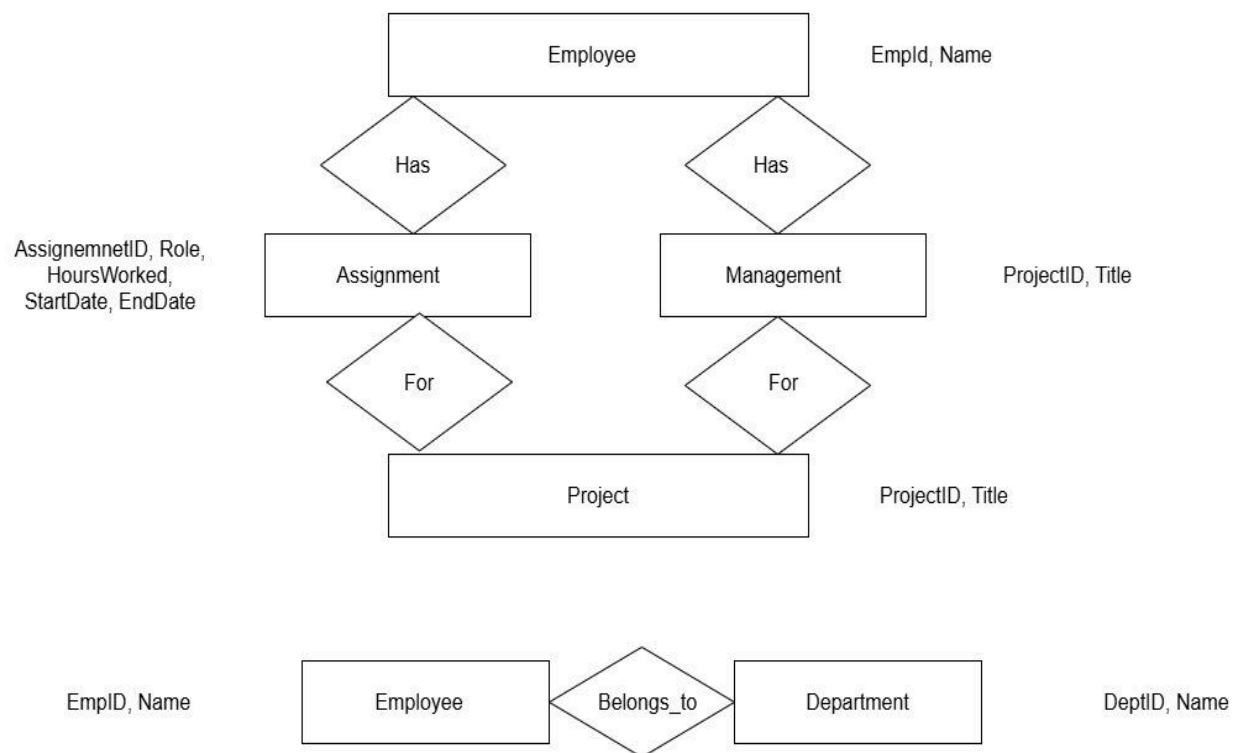
Corrected ER Diagram



The original ER model stored information at an inappropriate level because a customer can use a different shipping address for every order and the quantity of a product is specific to an order line. To correct this, order-specific delivery information was associated with the **Order**, while a new **OrderLine** entity was introduced to store the quantity of each product in a particular order. The shipment structure was also modified so that one order can be divided into multiple shipments and a single order line can be partially fulfilled from different warehouses. Separate shipment-line information allows the quantity shipped and warehouse used for each part of an order to be recorded, while shipment stores its own dispatch date and carrier information. These changes preserve historical order data, line-item quantities, split shipments, and partial fulfilment.

4. Project Staffing and Roles

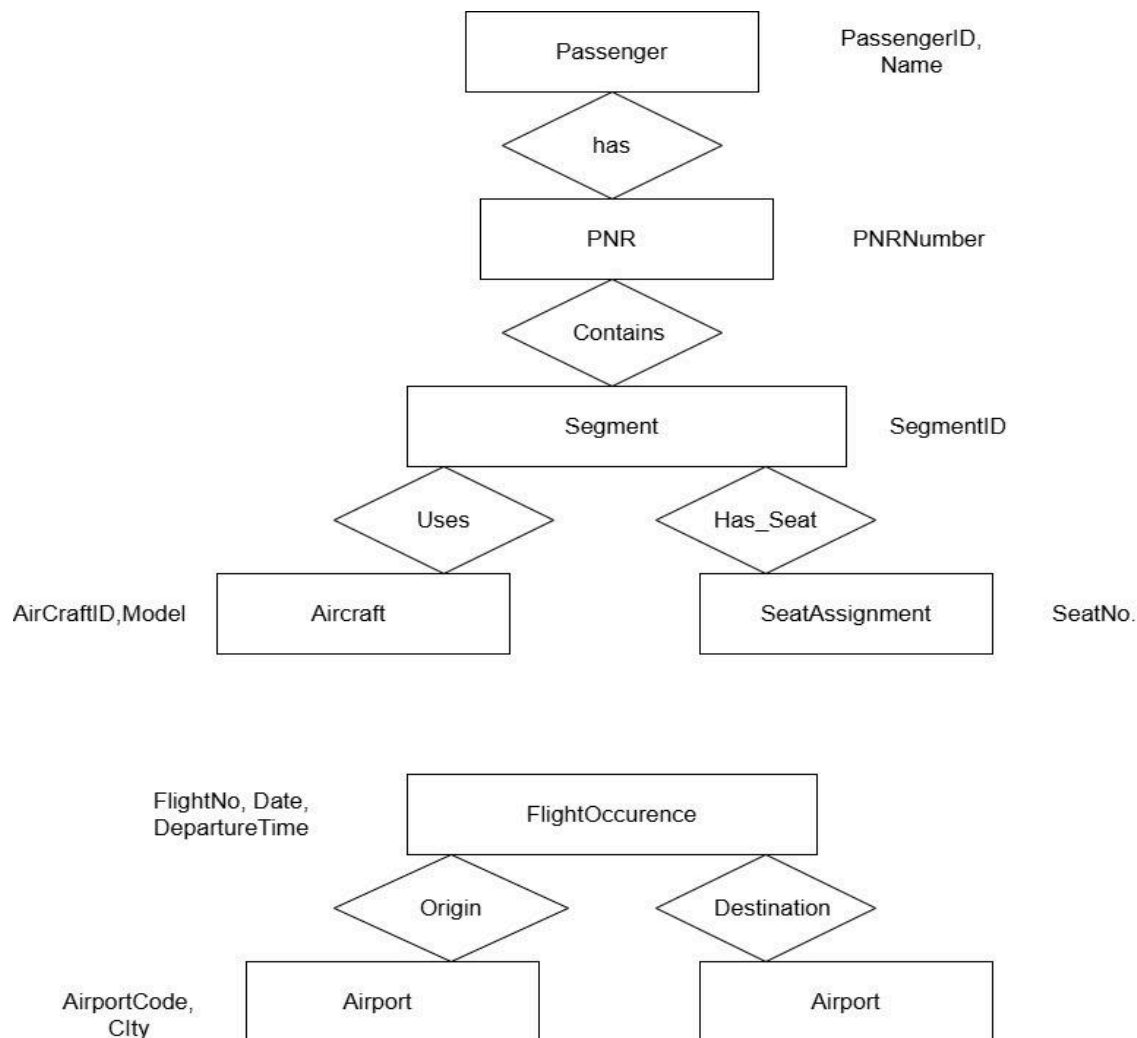
Corrected ER Diagram



The original ER model stored Role with Employee and HoursWorked with Project, which could not represent information that changes for a particular employee-project assignment or over time. To correct this, the employee's role was made assignment-specific so that the same employee can have different roles on different projects, while hours were associated with the employee-project assignment and its relevant time period. The model was also modified to represent project management as a time-dependent relationship, allowing different employees to manage the same project at different times. This preserves the history of employee assignments, roles, working hours, and project managers, including cases where an employee leaves and later rejoins the same project.

5. Project Staffing and Roles

Corrected ER Diagram



The original ER model treated Flight as the main entity for booking and aircraft assignment, which could not distinguish between different occurrences of the same flight number or represent a booking containing multiple passengers and flight segments. To correct this, a **Flight Occurrence** was introduced to represent a particular flight number on a specific date, and a **PNR/Booking** structure was introduced to group passengers and flight segments. Aircraft assignment was associated with the particular flight occurrence so that the aircraft can change for different occurrences. Seat assignment was also moved to the segment/flight-occurrence level because a passenger may have a different seat on different segments. These changes preserve PNR information, individual flight occurrences, aircraft changes, and segment-specific seat assignments.